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(54) **EXERCISE MACHINE**

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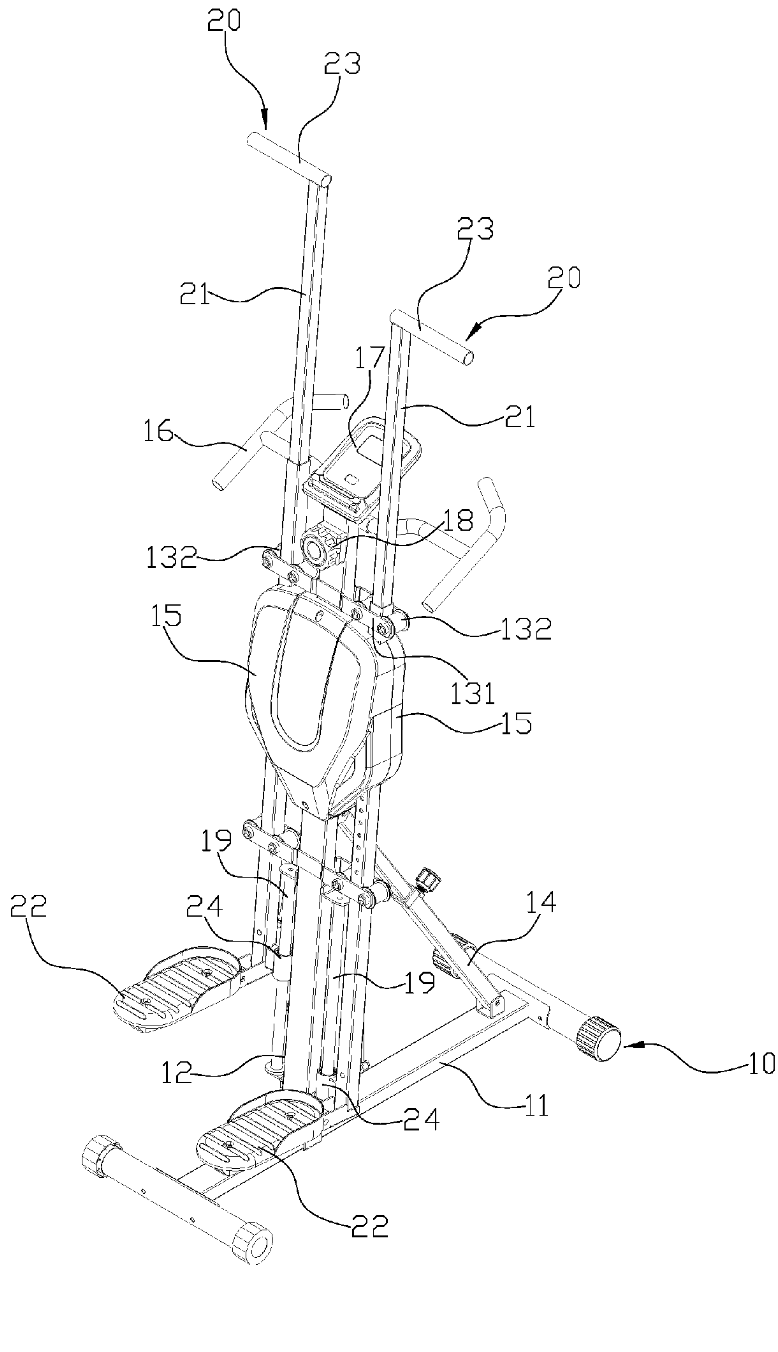
(57) **ABSTRACT**

An exercise machine includes a main body, the two actuators, a traction device and a resistance unit. The main body has a base, and a supporting frame standing on the base, with two sides of the supporting frame each has an assembling space. The actuator has a pole with a pedal at a lower end and a handle at a upper end. The traction device has a sprocket and a central axis, and the sprocket has a chain, and the resistance unit has a flywheel and a magnetic base.



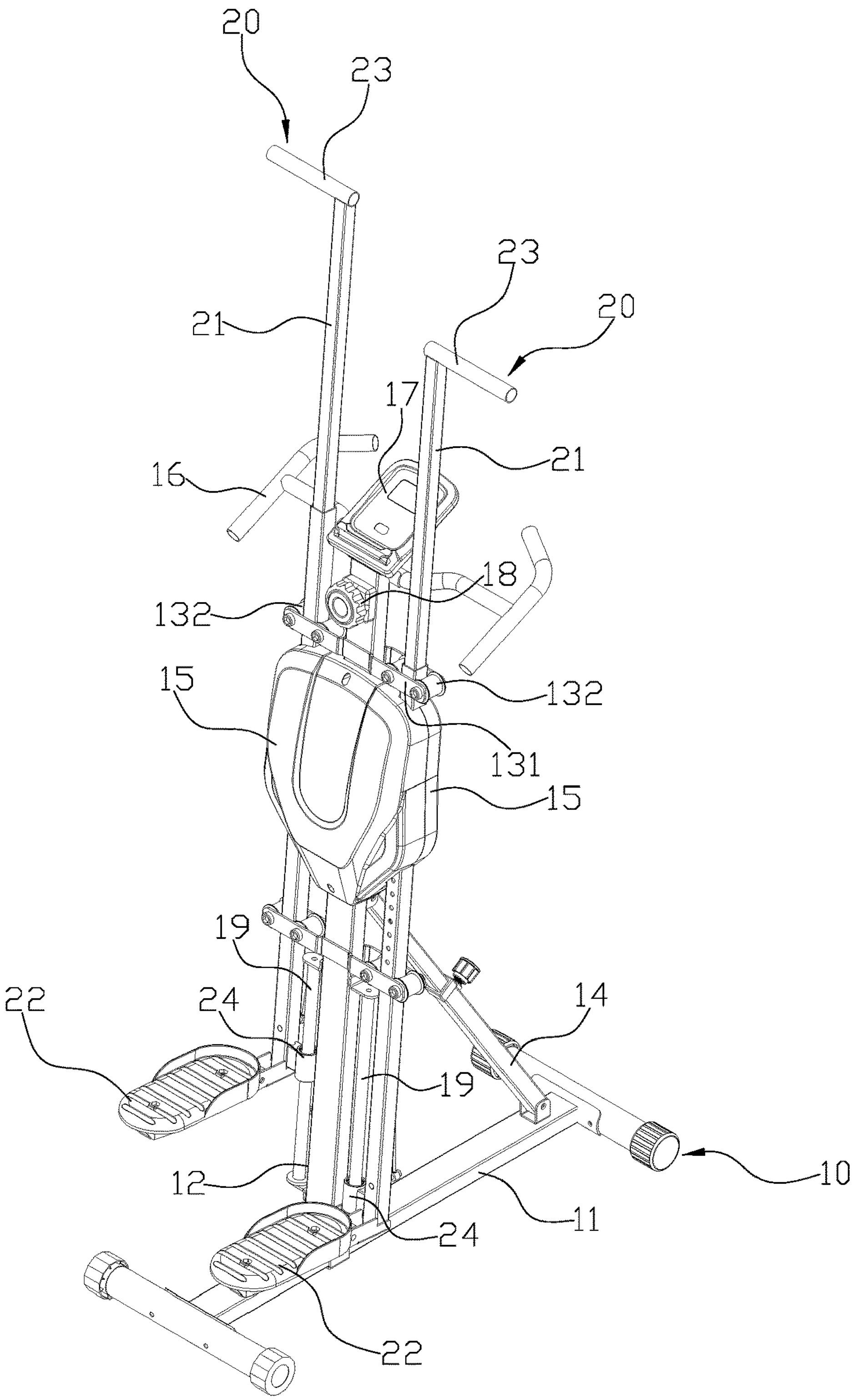

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FIG.1

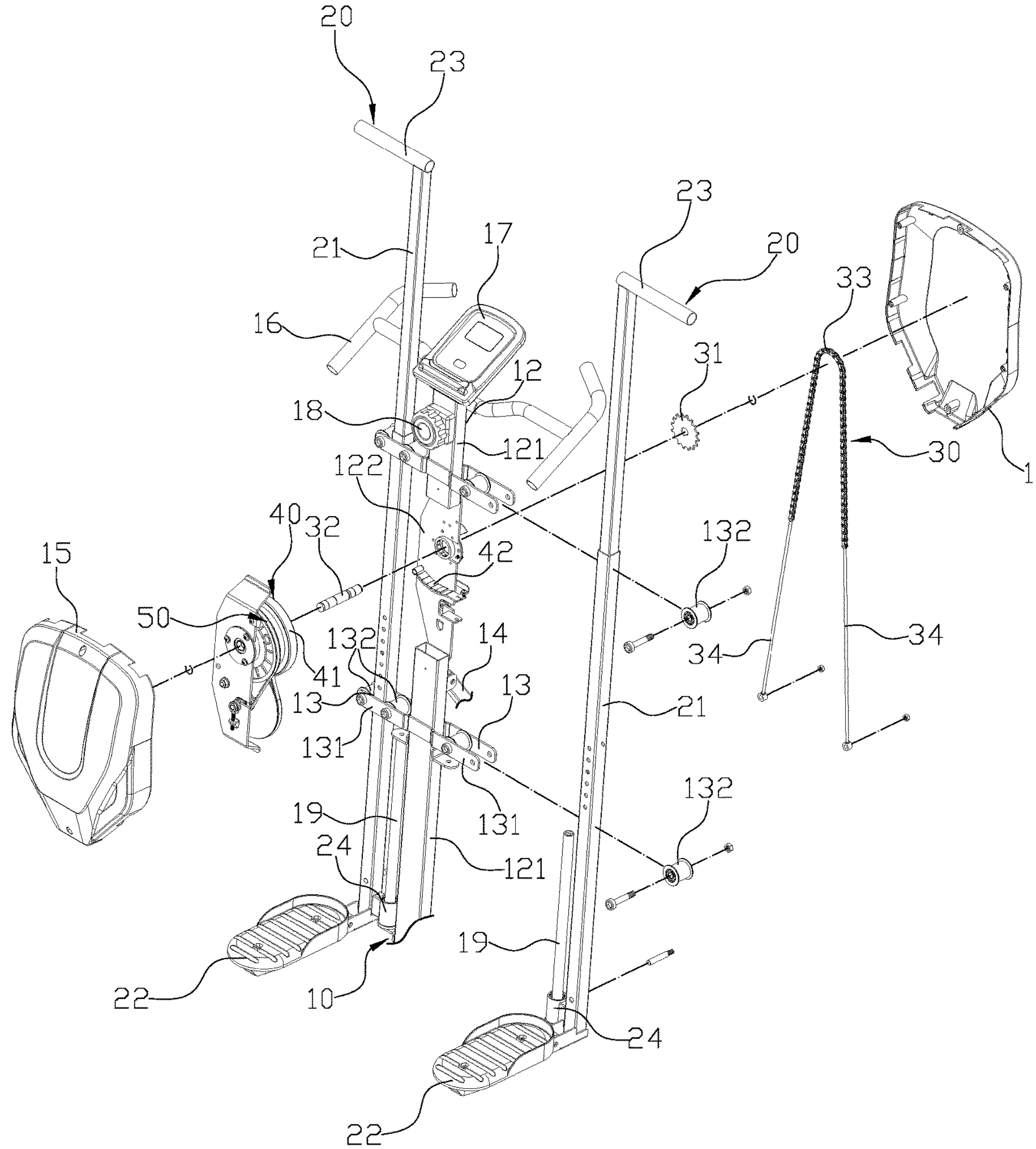
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FIG.2

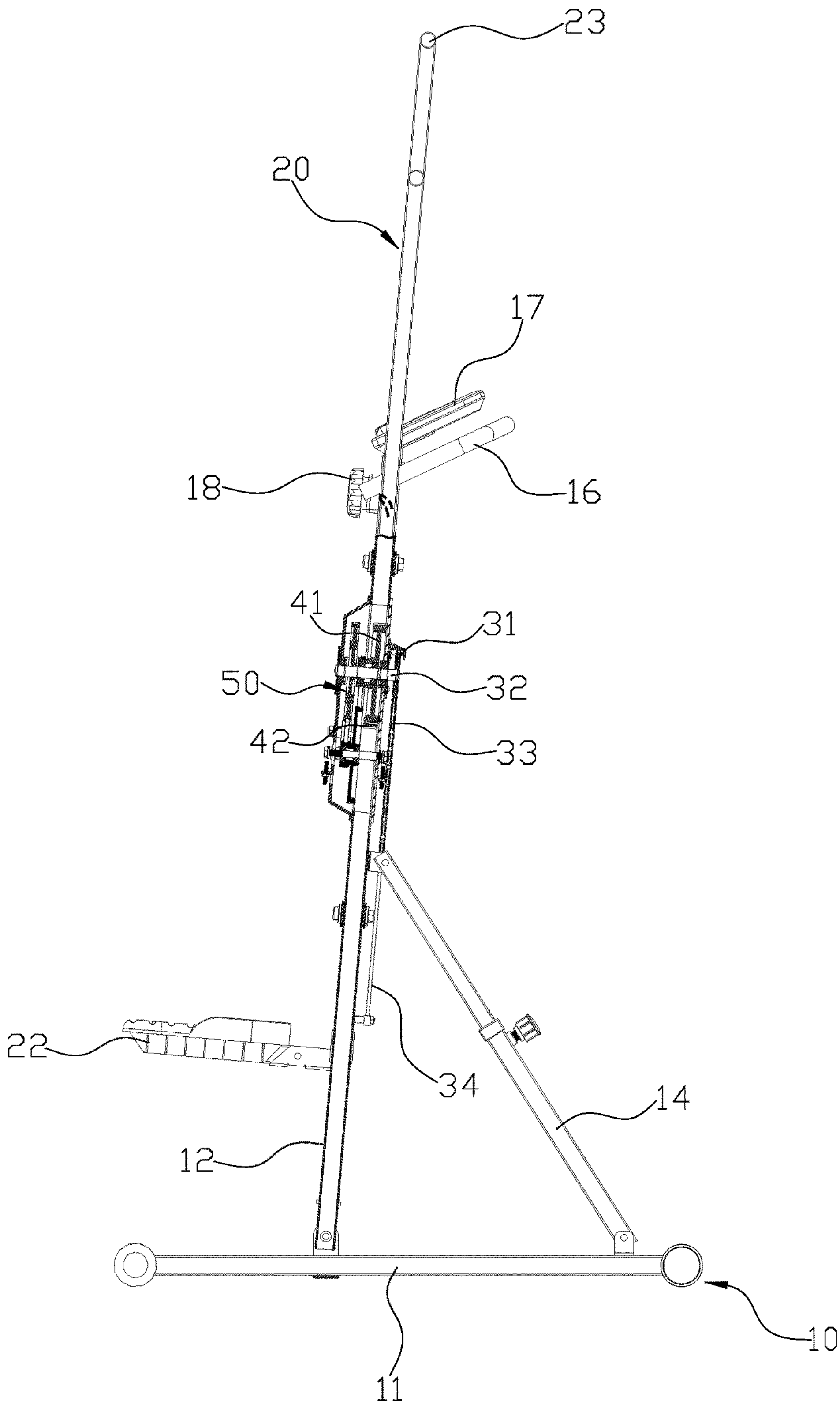

FIG.3

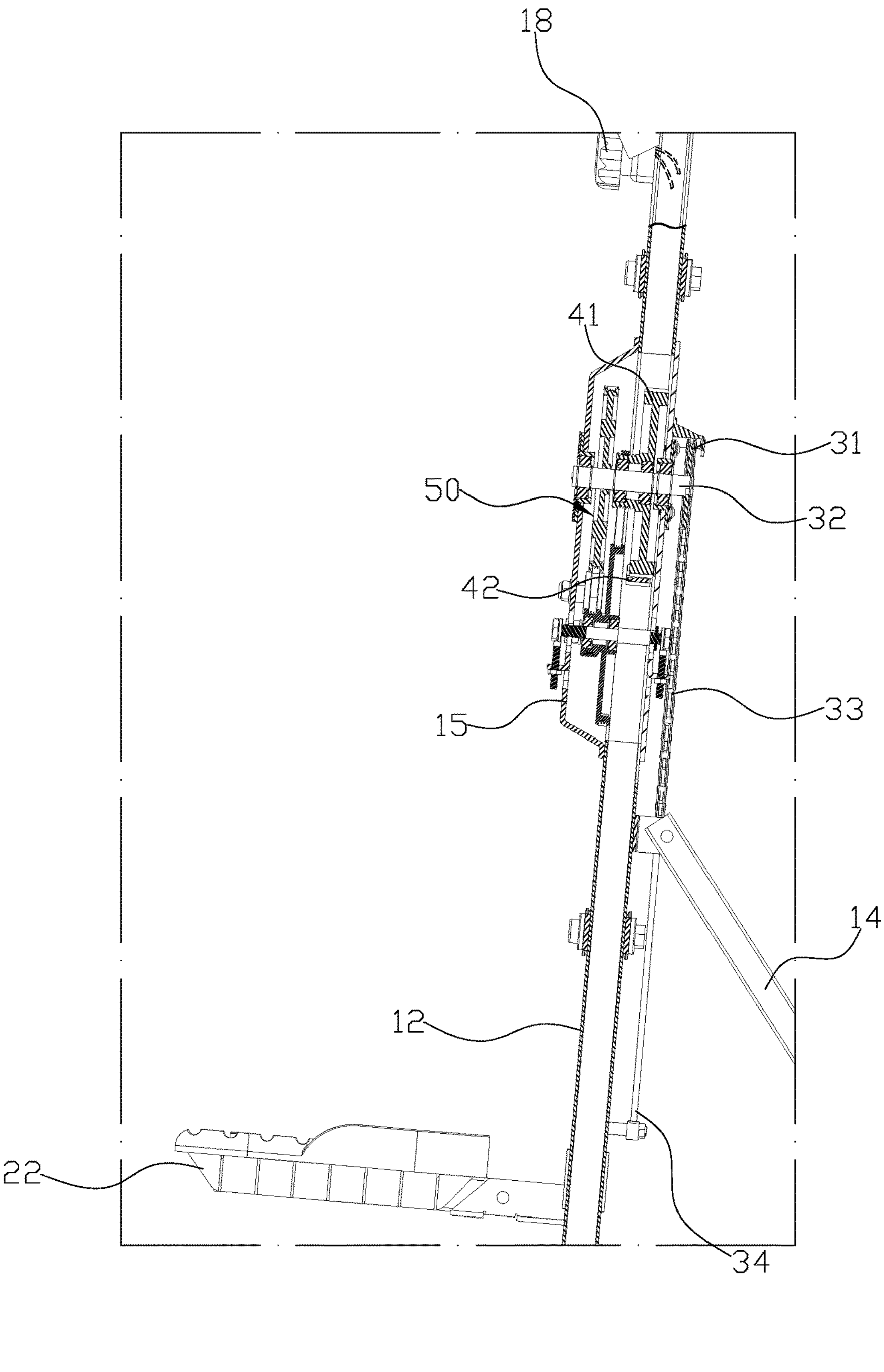

FIG.4

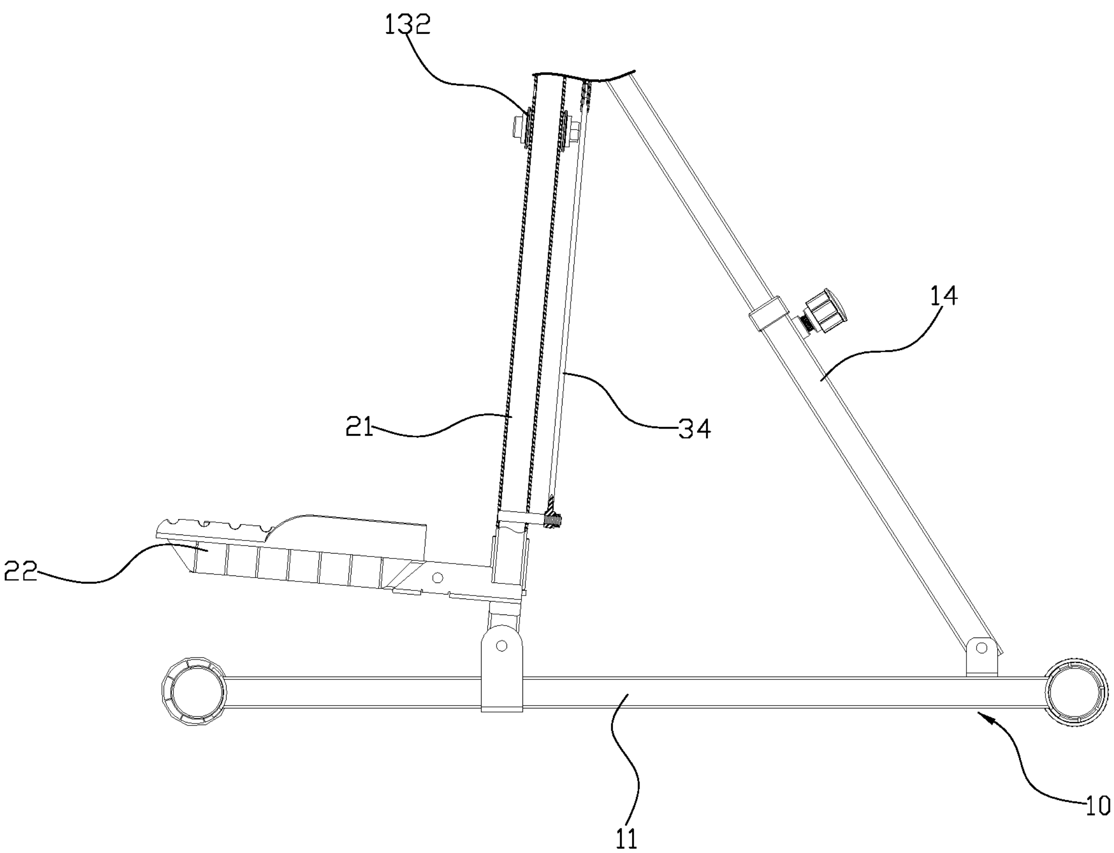

FIG.5

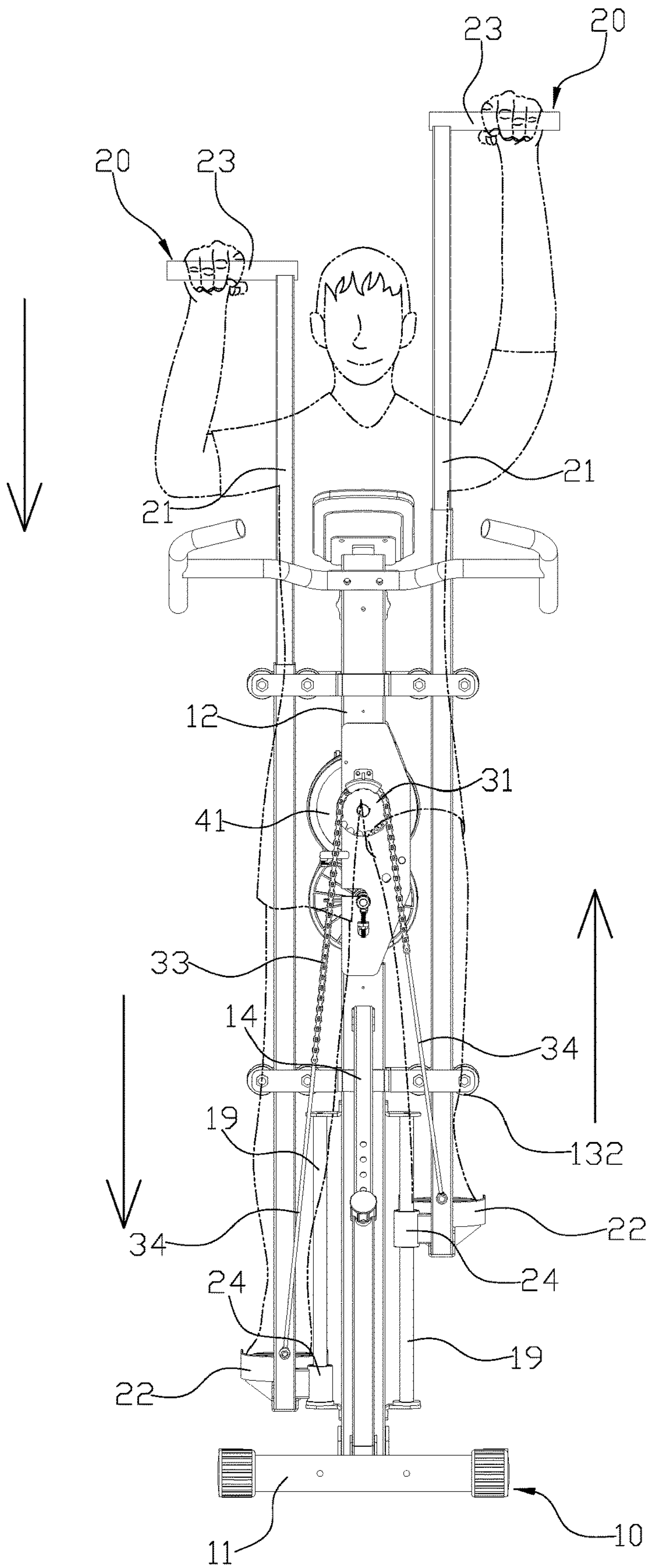

FIG.6

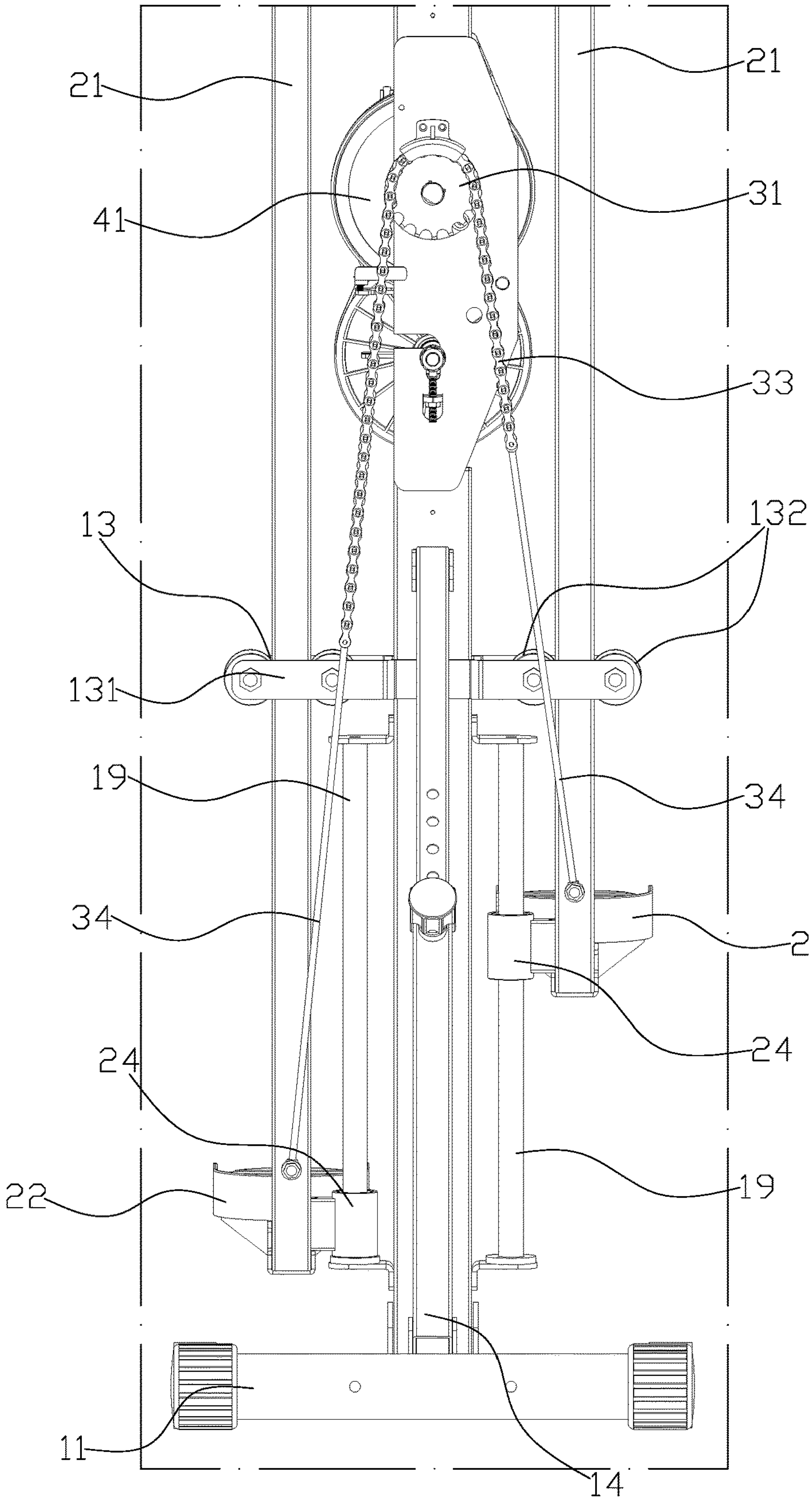

FIG.7

EXERCISE MACHINE

BACKGROUND OF INVENTION

Field of Invention

[0001] The present invention relates to an exercise machine.

Description of the Related Art

[0002] Modern life people are busy with work and live at a fast pace. The focus of their life is mainly on work and family, and they rarely have time to exercise. As a result, weight gain and poor health are the common problems. However, going to the gym for exercise, time and space might also become issues. Therefore, many home exercise machines such as bikes, the flywheel, treadmills, and sliders . . . etc. various fitness equipment are developed.

[0003] However, the above conventional structure still has the following problems: when the exercise bike, treadmill, or sliding machine is operated, the user's arms can only make slight swings at the sides of the body. Therefore, naturally there is no way to use these fitness devices to help the arms to fully stretch the limbs. Fitness equipment that only focuses on the lower body exercise obviously cannot achieve full-body exercise, so the fitness effect is really limited, and it cannot play a good role in the prevention and improvement of frozen shoulder, arthritis and other upper limb issues.

[0004] Therefore, it is desirable to provide an exercise machine to mitigate and/or obviate the aforementioned problems.

SUMMARY OF THE INVENTION

[0005] An objective of present invention is to provide a sprayer gun connecting structure which is capable of improving the above-mention problems.

[0006] In order to achieve the above mentioned objective, an exercise machine has a main body, two actuators, a traction device, and a resistance unit. The main body has a base and a supporting frame, two sides of the supporting frame each having an assembling space for respectively accepting the two actuators. Each actuator has a pole with a pedal at a lower end and a handle at a higher end and the pole is inserted in one of the assembling spaces. The traction device has a sprocket and a central axis fixed with each other, the central axis disposed through the supporting frame and the sprocket pivoted behind the supporting frame, the sprocket mounted with a chain, and two ends of the chain are respectively connected to the two poles. The resistance unit has a flywheel attached to the central axis and a magnetic base pivoted on the supporting frame, wherein when the magnetic base is pushed close to or away from the flywheel, the flywheel generates a different resistance to the actuators.

[0007] Other objects, advantages, and novel features of invention will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. **1** is a three-dimensional combination view of a preferred embodiment of the present invention.

[0009] FIG. **2** is a three-dimensional exploded view of the preferred embodiment of the present invention.

[0010] FIG. **3** is a combined cross-sectional view of the preferred embodiment of the present invention.

[0011] FIG. **4** is a partial enlarged view of the combined section of the preferred embodiment of the present invention.

[0012] FIG. **5** is another partial enlarged view of the combined cross-section of the preferred embodiment of the present invention.

[0013] FIG. **6** shows a fitness state of the preferred embodiment of the present invention.

[0014] FIG. **7** is a partial enlarged view of the actuator being raised and lowered interactively according to the preferred embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0015] First, please refer to FIG. **1** to FIG. **5**. An exercise machine comprises a main body **10**, the two actuator **20**, a traction device **30** and a resistance unit **40**. The main body **10** has a base **11** and a supporting frame **12**, and two sides of the supporting frame **12** each having an assembling space **13** for respectively accepting the two actuators **20**. The actuator **20** has a pole **21** with a pedal **22** at a lower end and a handle **23** at a higher end and the pole **21** is inserted in one of the assembling spaces **13** such that the pedal **22** and the handle **23** can be driven up and down by the up and down displacement of the pole **21** in the assembling space **13**. The traction device **30** has a sprocket **31** and a central axis **32** fixed with each other, the central axis **32** is disposed through the supporting frame **12** and the sprocket **31** is pivoted behind the supporting frame **12**. The sprocket **31** is mounted with a chain **33**, and two ends of the chain **33** are respectively connected to the two poles **21**. Therefore, when a pedal **22** is stepped down, the traction device **30** pulls up another actuator **20** to achieve a continuous pedaling cycle during exercise. The resistance unit **40** has a flywheel **41** attached to the central axis **32** and a magnetic base **42** pivoted on the supporting frame **12**, wherein when the magnetic base **42** is pushed close to or away from the flywheel **41**, the flywheel **41** generates a different resistance to the actuators **20**.

[0016] Moreover, the supporting frame **12** is pivoted on the base **11**, two ends of a telescopic pole **14** are respectively pivoted on a back of the supporting frame **12** and the base **11**, which provides diagonal support to the supporting frame **12** and allows the supporting frame **12** to be capable of having different standing angles and to be foldable.

[0017] Furthermore, the supporting frame **12** comprises two rods **121** and a plate **122** disposed between the two rods **121**, and the traction device **30** and the resistance unit **40** are both disposed on the plate **122**.

[0018] In addition, a front and rear sides of the plate **122** are each provided with a shell **15**, so that the two shells **15** are combined to cover the resistance unit **40**, the sprocket **31** and the central axis **32**.

[0019] Also, the assembling space **13** is formed by a positioning frame **131** of the supporting frame **12**, and the positioning frame **131** comprises at least two rollers **132** for moving the pole **21**.

[0020] Additionally, the supporting frame **12** further comprises an armrest **16** and a control panel **17**.

[0021] Also, the supporting frame **12** is provided with a resistance controller **18**. The resistance controller **18** is connected to the magnetic base **42** through a cable, so that through the rotation of the resistance controller **18**, the magnetic base **42** can be driven close to or away from the flywheel **41** by the cable.

[0022] Furthermore, two sides of the supporting frame **12** respectively have a guiding column **19**, a side of each of the two poles **21** having a guiding sleeve **24** for accepting the guiding column **19**, which can limit the maximum travel distance of the actuator **20**.

[0023] Also, a length of the pole **21** is adjustable.

[0024] Moreover, two ends of the chain **33** are respectively connected to a cable **34** such that the chain **33** can be connected to the two pole **21** by the cables **34**. And, the cable **34** is a steel cable.

[0025] In addition, the central axis **32** further comprises a pulley set **50** for providing a predetermined rotation speed ratio.

[0026] Please refer to FIGS. **6** and **7**. For actual operation, a user places his or her two feet on the two pedals **22** and steps the pedals **22** alternately, so that the two arms holding the two handles **23** can be lifted up alternately, and through the complete extension of the limbs, whole-body movements so the with exercise effect are achieved. By fully lifting the arms, it can also prevent and improve symptoms such as frozen shoulder, arthritis, etc. by using the hands and feet together to produce a similar climbing effect, the exercise posture of the exercise machine of the present invention also provides operating pleasure that is different from other fitness equipment.

[0027] Although the present invention has been explained in relation to its preferred embodiment, it is to be understood that many other possible modifications and variations can be made without departing from the spirit and scope of invention as hereinafter claimed.

What is claimed is:

1. An exercise machine comprising a main body, two actuators, a traction device, and a resistance unit, wherein:

the main body comprises a base and a supporting frame, two sides of the supporting frame each having an assembling space for respectively accepting the two actuators;

each actuator comprises a pole with a pedal at a lower end and a handle at a higher end and the pole is inserted in one of the assembling spaces;

the traction device comprises a sprocket and a central axis fixed with each other, the central axis disposed through the supporting frame and the sprocket pivoted behind the supporting frame, the sprocket mounted with a chain, and two ends of the chain are respectively connected to the two poles; and

the resistance unit comprises a flywheel attached to the central axis and a magnetic base pivoted on the supporting frame, wherein when the magnetic base is pushed close to or away from the flywheel, the flywheel generates a different resistance to the actuators.

2. The exercise machine as claimed in claim **1**, wherein the supporting frame is pivoted on the base, two ends of a telescopic pole are respectively pivoted on a back of the supporting frame and the base, which provides diagonal support to the supporting frame and allows the supporting frame to be capable of having different standing angles and to be foldable.

3. The exercise machine as claimed in claim **1**, wherein the supporting frame comprises two rods and a plate disposed between the two rods, and the traction device and the resistance unit both disposed on the plate.

4. The exercise machine as claimed in claim **1**, wherein the assembling space is formed by a positioning frame of the supporting frame, and the positioning frame comprises at least two rollers for moving the pole.

5. The exercise machine as claimed in claim **1**, wherein the supporting frame further comprises an armrest and a control panel.

6. The exercise machine as claimed in claim **1**, wherein two sides of the supporting frame respectively have a guiding column, a side of each of the two poles having a guiding sleeve for accepting the guiding column.

7. The exercise machine as claimed in claim **1**, wherein a length of the pole is adjustable.

8. The exercise machine as claimed in claim **1**, wherein two ends of the chain are respectively connected to a cable such that the chain can be connected to the two pole by the cables.

9. The exercise machine as claimed in claim **8**, wherein the cable is a steel cable.

10. The exercise machine as claimed in claim **1**, wherein the central axis further comprises a pulley set for providing a predetermined rotation speed ratio.

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